UTM UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

Semester II 2019/2020

Subject : Bioinformatics I (SCSB2103)
Section : 01 – Dr Haslina Hashim
Topic : Lab 05 – Advanced Database Searching

Koo Xuan (A23CS0300)

1) Create an artificial protein sequence consisting of human RBP4 followed by the C2 domain of human protein kinase Cα. An example of this is shown in Web Document 5.5 (available in your e-Learning site). Enter this combined sequence into a PSI-BLAST search

   i) In general, are multiple domains always detected by the PSI-BLAST program?

   ii) Do any naturally occurring proteins have both lipocalin and C2 domains?

First, find RBP4 Homo sapiens and select NP_006735.2 the copy the FASTA sequence.

>NP_006735.2 retinol-binding protein 4 isoform a precursor [Homo sapiens]

MKWVWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFSVDETGQ

MSATAKGRVRLLNNWDVCADMVGTFTDTEDPAKFKMKYWGVASFLQKGNDDHWIVDTDYDTYAVQYSCRL

LNLDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLL

retinol-binding protein 4 isoform a precursor [Homo sapiens]

NCBI Reference Sequence: NP_006735.2

GenPept Identical Proteins Graphics

```
>NP_006735.2 retinol-binding protein 4 isoform a precursor [Homo sapiens]
MKWVWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFSVDETGQ
MSATAKGRVRLLNNWDVCADMVGTFTDTEDPAKFKMKYWGVASFLQKGNDDHWIVDTDYDTYAVQYSCRL
LNLDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLL
```

Then find Protein kinase C alpha Homo sapiens and select NP_002728.2 the copy the FASTA sequence.

>NP_002728.1 protein kinase C alpha type [Homo sapiens]

MADVFPGNDSTASQDVANRFARKGALRQKNVHEVKDHKFIARFFKQPTFCSHCTDFIWGFGKQGFQCQVC

CFVVHKRCHEFVTFSCPGADKGPDTDDPRSKHKFKIHTYGSPTFCDHCGSLLYGLIHQGMKCDTCDMNVH

KQCVINVPSLCGMDHTEKRGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQ

KTKTIRSTLNPQWNESFTFKLKPSDKDRRLSVEIWDWDRTTRNDFMGSLSFGVSELMKMPASGWYKLLNQ

EEGEYYNVPIPEGDEEGNMELRQKFEKAKLGPAGNKVISPSEDRKQPSNNLDRVKLTDFNFLMVLGKGSF

GKVMLADRKGTEELYAIKILKKDVVIQDDDVECTMVEKRVLALLDKPPFLTQLHSCFQTVDRLYFVMEYV

NGGDLMYHIQQVGKFKEPQAVFYAAEISIGLFFLHKRGIIYRDLKLDNVMLDSEGHIKIADFGMCKEHMM

DGVTTRTFCGTPDYIAPEIIAYQPYGKSVDWWAYGVLLYEMLAGQPPFDGEDEDELFQSIMEHNVSYPKS

LSKEAVSICKGLMTKHPAKRLGCGPEGERDVREHAFFRRIDWEKLENREIQPPFKPKVCGKGAENFDKFF

TRGQPVLTPPDQLVIANIDQSDFEGFSYVNPQFVHPILQSAV

FASTA ▾

protein kinase C alpha type [Homo sapiens]

NCBI Reference Sequence: NP_002728.1

This sequence has been updated. See current version.

GenPept Identical Proteins Graphics


```
>NP_002728.1 protein kinase C alpha type [Homo sapiens]
MADVFPGNDSTASQDVANRFARKGALRQKNVHEVKDHKFIARFFKQPTFCSHCTDFIWGFGKQGFQCQVC
CFVVHKRCHEFVTFSCPGADKGPDTDDPRSKHKFKIHTYGSPTFCDHCGSLLYGLIHQGMKCDTCDMNVH
KQCVINVPSLCGMDHTEKRGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQ
KTKTIRSTLNPQWNESFTFKLKPSDKDRRLSVEIWDWDRTTRNDFMGSLSFGVSELMKMPASGWYKLLNQ
EEGEYYNVPIPEGDEEGNMELRQKFEKAKLGPAGNKVISPSEDRKQPSNNLDRVKLTDFNFLMVLGKGSF
GKVMLADRKGTEELYAIKILKKDVVIQDDDVECTMVEKRVLALLDKPPFLTQLHSCFQTVDRLYFVMEYV
NGGDLMYHIQQVGKFKEPQAVFYAAEISIGLFFLHKRGIIYRDLKLDNVMLDSEGHIKIADFGMCKEHMM
DGVTTRTFCGTPDYIAPEIIAYQPYGKSVDWWAYGVLLYEMLAGQPPFDGEDEDELFQSIMEHNVSYPKS
LSKEAVSICKGLMTKHPAKRLGCGPEGERDVREHAFFRRIDWEKLENREIQPPFKPKVCGKGAENFDKFF
TRGQPVLTPPDQLVIANIDQSDFEGFSYVNPQFVHPILQSAV
```


Here is the C2 domain of PKCA, obtained by clicking the C2 “region” link:

>NP_002728.1:159-289 protein kinase C alpha type [Homo sapiens]

RGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQKTKTIRSTLNPQWNESFT

FKLKPSDKDRRLSVEIWDWDRTTRNDFMGSLSFGVSELMKMPASGWYKLLNQEEGEYYNVP

FASTA

protein kinase C alpha type [Homo sapiens]

NCBI Reference Sequence: NP_002728.1

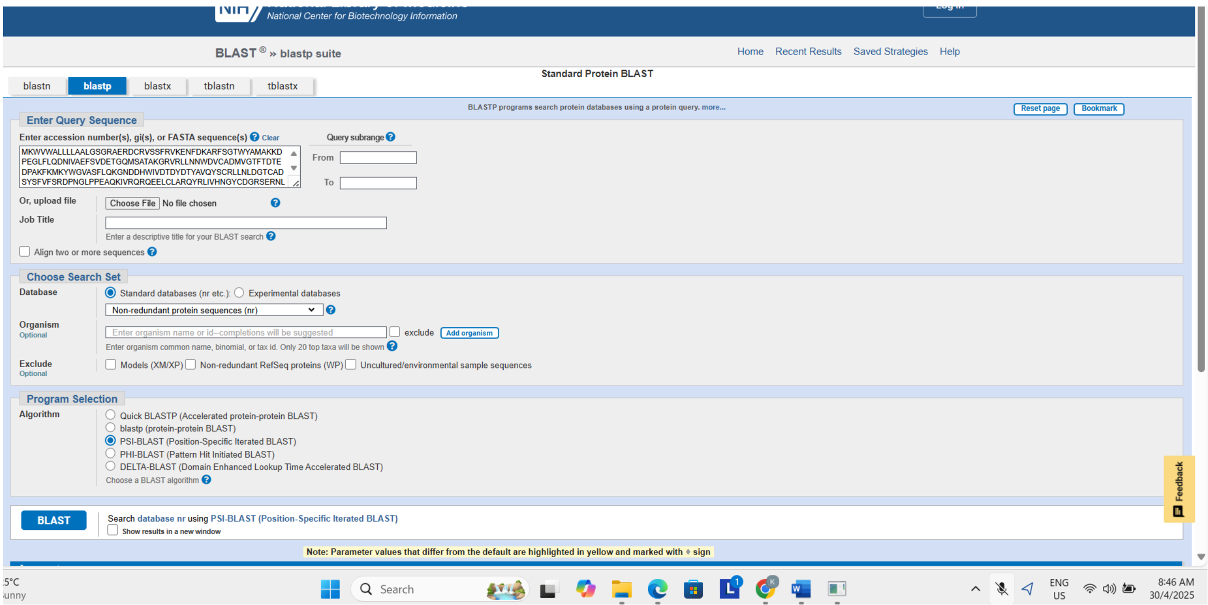
This sequence has been updated. See current version.

GenPept Identical Proteins Graphics

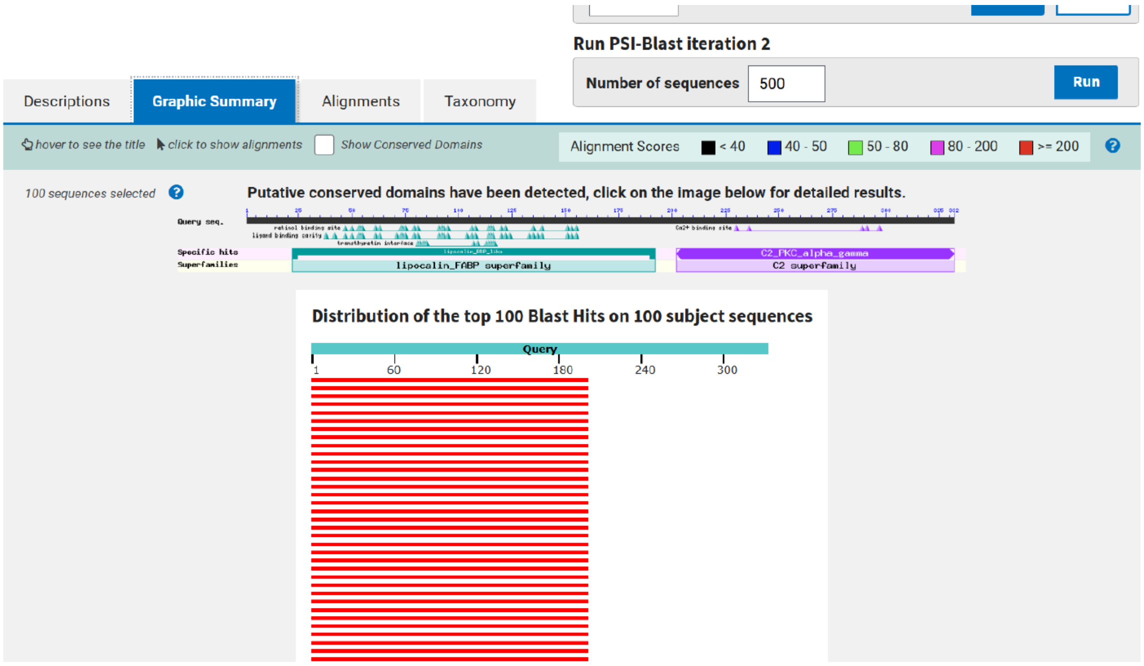
```
>NP_002728.1:159-289 protein kinase C alpha type [Homo sapiens]
RGRIYLKAEVADEKLHVTVRDAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQKTKTIRSTLNPQWNESFT
FKLKPSDKDRRLSVEIWDWDRTTRNDFMGSLSFGVSELMKMPASGWYKLLNQEEGEYYNVP
```

Here is a chimeric protein:

MKWVWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIVAEFSVDETGQMSA
TAKGRVRLLNNWDVCADMVGTFTDTEDPAKFKMKYWGVASFLQKGNDDHWIVDTDYDTYAVQYSCRLLNLDG
TCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLARQYRLIVHNGYCDGRSERNLLRGRIYLKAEVADEKLHVTVR
DAKNLIPMDPNGLSDPYVKLKLIPDPKNESKQKTKTIRSTLNPQWNESFTFKLKPSDKDRRLSVEIWDWDRTTRND
FMGSLSFGVSELMKMPASGWYKLLNQEEGEYYNVP



This shows the PSI-BLAST result:



The graphical portion of the result suggests that there are matches to either RBP4 or to PKC but not both; the RBP4 portion of the protein is larger so it accumulates higher scores (and lower E values) and those results are listed first. By inspection there are no proteins with both domains.

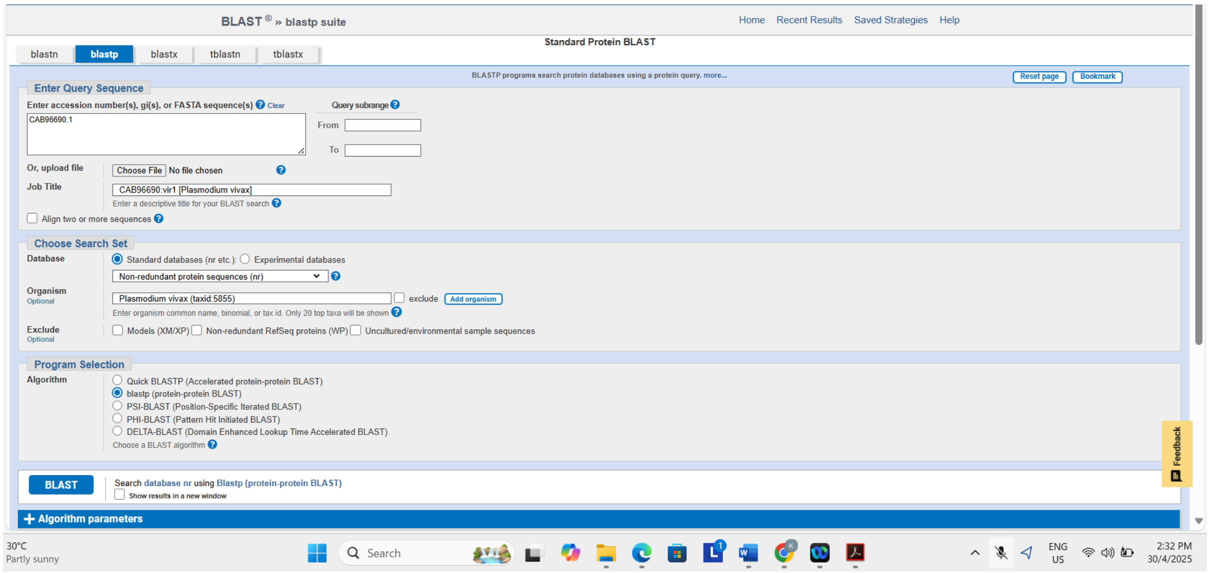
2) The purpose of this problem is to compare BLASTP to DELTA-BLAST

The malarium parasite Plasmodium vivax has a multigene family called vir that is specific to that organism (del Portillo et al., 2001). There are 600–1000 copies of these genes, and they may have a role in causing chronic infection through antigenic variation.

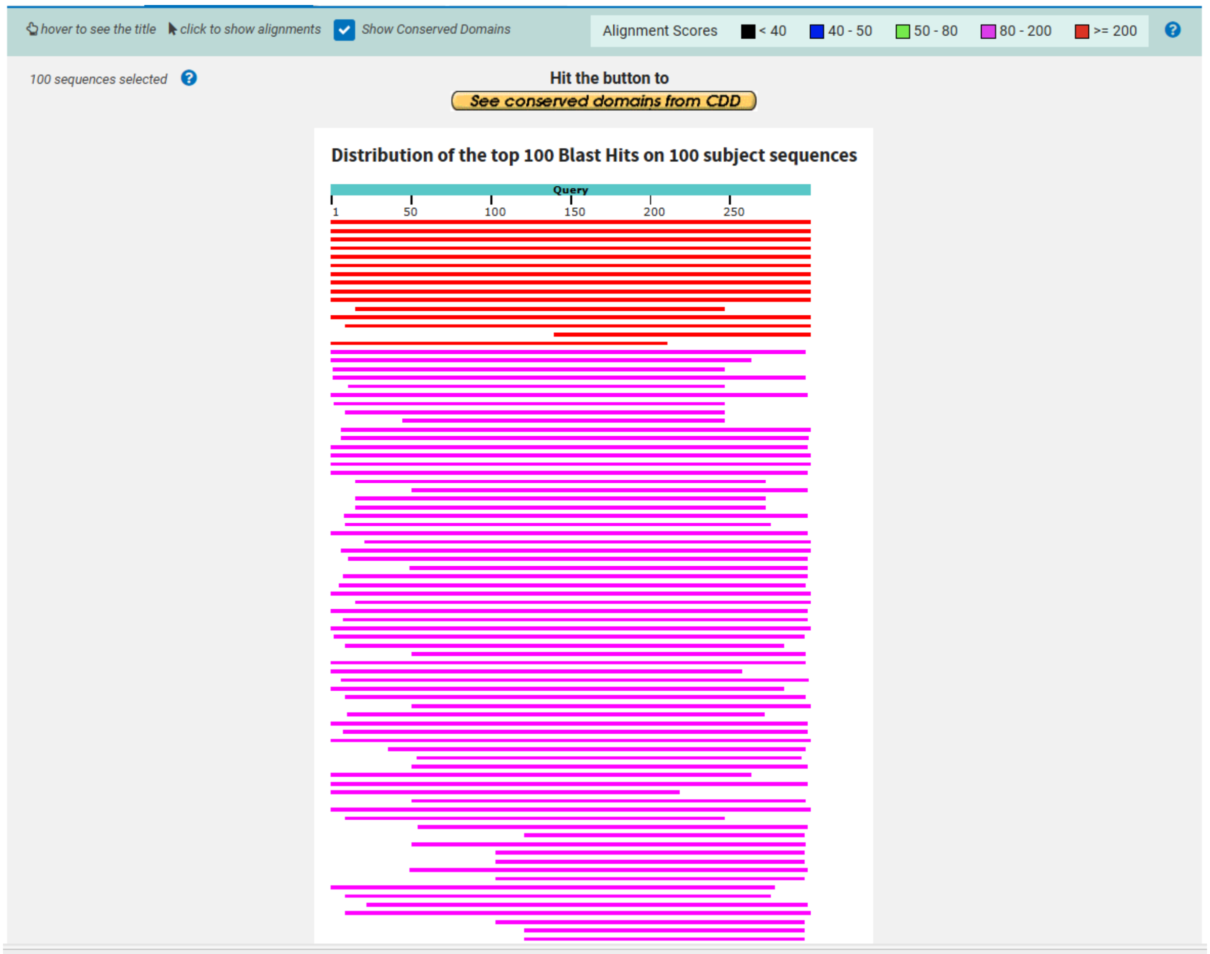
i) Select vir1 and perform a BLASTP search of the nonredundant protein database (restricting the species to Plasmodium vivax).

ii) Then perform a DELTA-BLAST search with the same entry. For each search, approximately how many proteins have an E value less than 1e-10?

First , find vir-1 protein from Plasmodium vivax search NCBI Protein: "Plasmodium vivax"[ORGN] vir1 , then enter its query sequence CAB96690.1



Here is the BLASTP result:



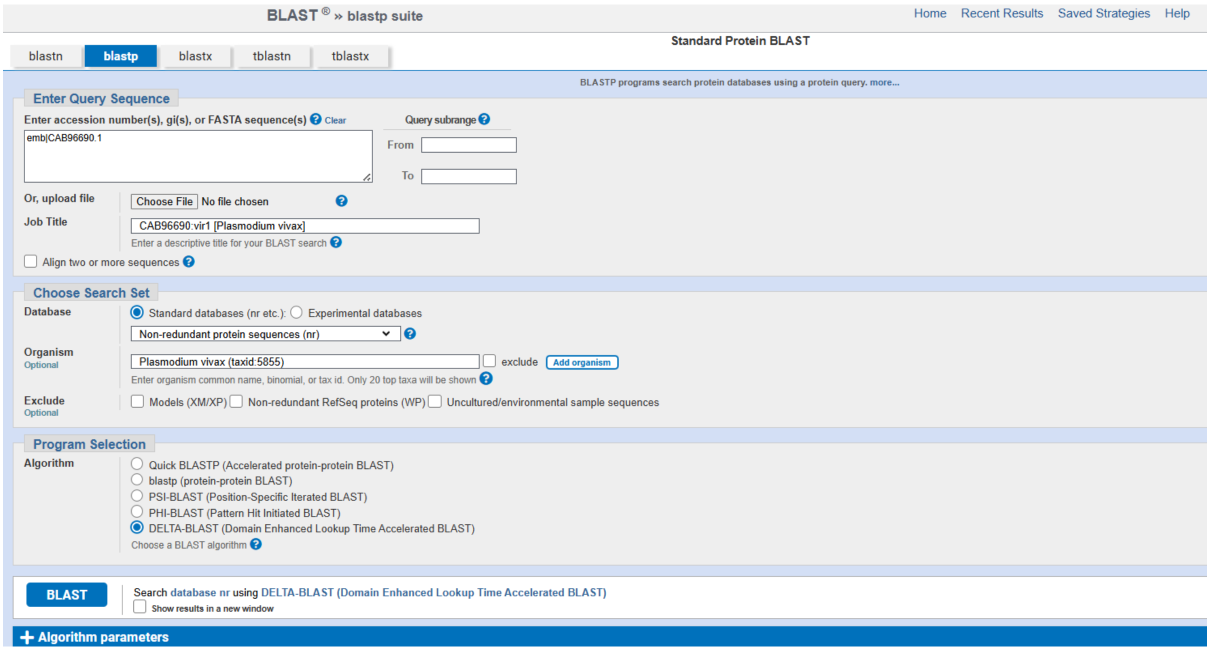
Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

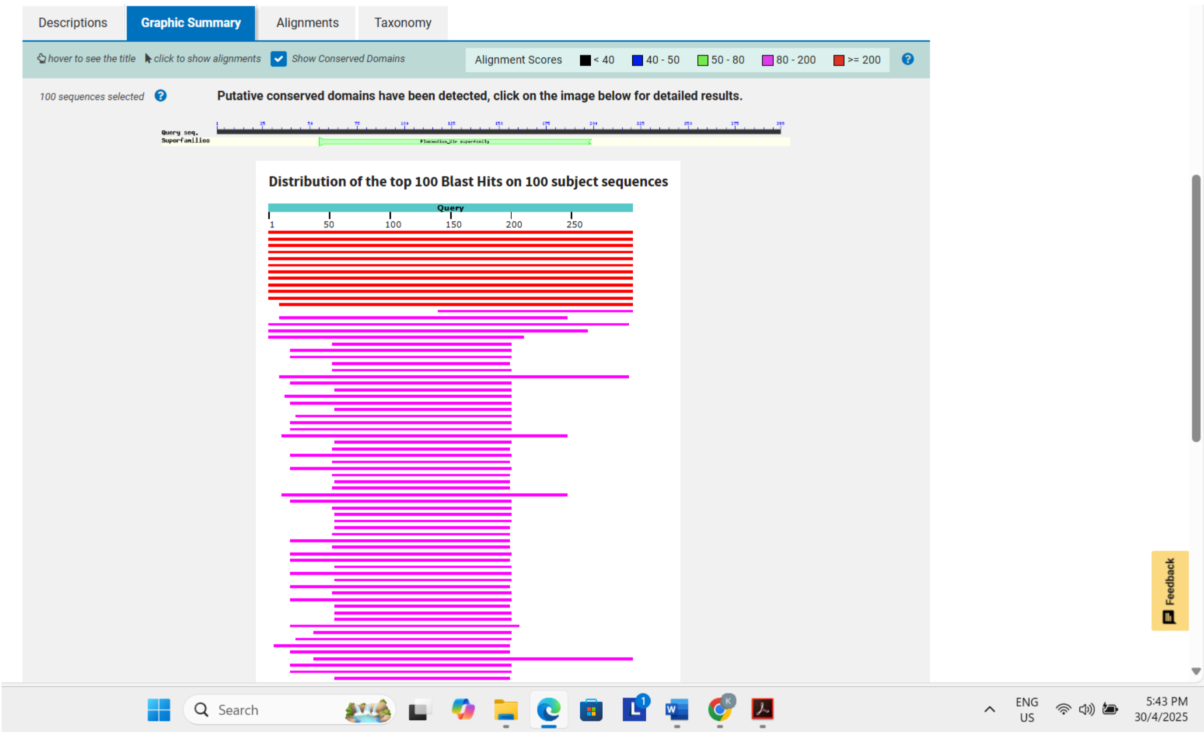
select all 100 sequences selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

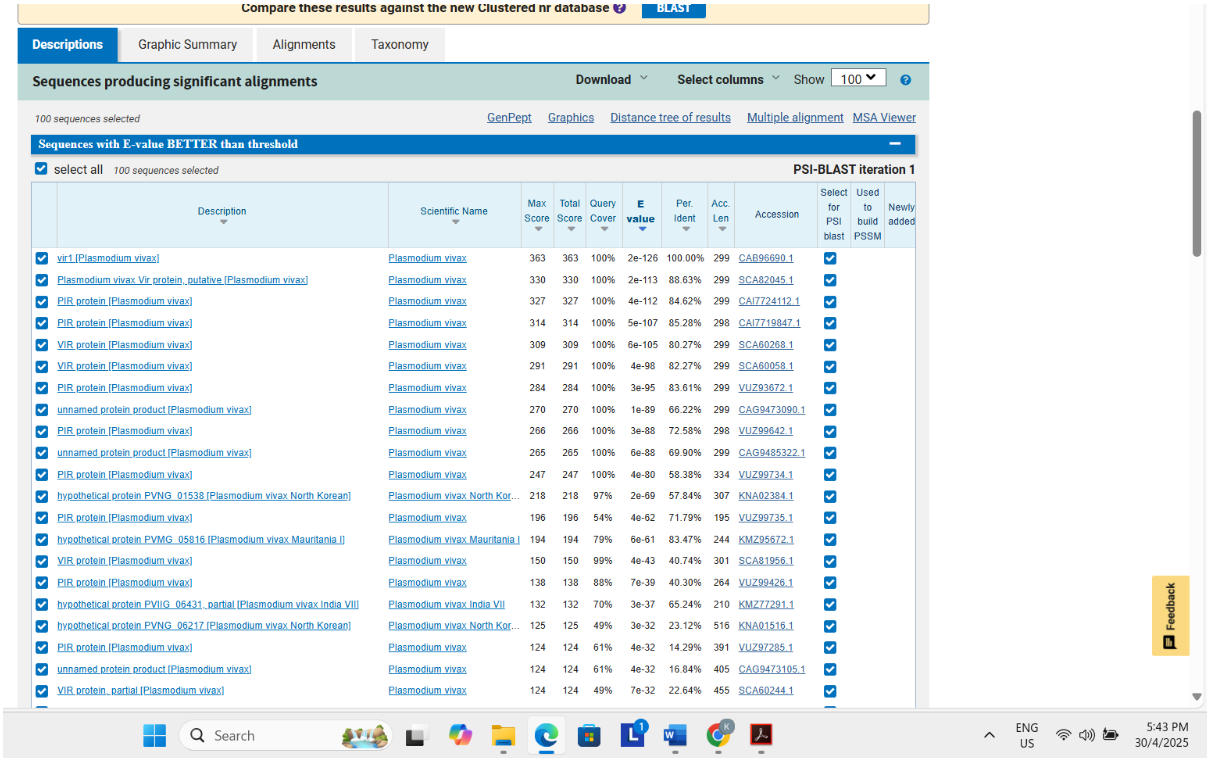
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
vir1 [Plasmodium vivax]	Plasmodium vivax	607	607	100%	0.0	100.00%	299	CAB96690.1
Plasmodium vivax Vir protein, putative [Plasmodium vivax]	Plasmodium vivax	535	535	100%	0.0	88.63%	299	SCA82045.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	516	516	100%	0.0	84.62%	299	CAI7724112.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	510	510	100%	0.0	85.28%	298	CAI7719847.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	494	494	100%	3e-178	83.61%	299	VUZ93672.1
VIR protein [Plasmodium vivax]	Plasmodium vivax	488	488	100%	8e-176	82.27%	299	SCA60058.1
VIR protein [Plasmodium vivax]	Plasmodium vivax	481	481	100%	4e-173	80.27%	299	SCA60268.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	423	423	100%	4e-150	72.58%	298	VUZ99642.1
unnamed protein product [Plasmodium vivax]	Plasmodium vivax	416	416	100%	3e-147	69.90%	299	CAG9485322.1
unnamed protein product [Plasmodium vivax]	Plasmodium vivax	396	396	100%	1e-139	66.22%	299	CAG9473090.1
hypothetical protein PVMG_05816 [Plasmodium vivax Mauritania I]	Plasmodium vivax Mauritania I	389	389	77%	2e-137	84.78%	244	KMZ95672.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	377	377	100%	3e-131	58.38%	334	VUZ99734.1
hypothetical protein PVNG_01538 [Plasmodium vivax North Korean]	Plasmodium vivax North Korean	345	345	97%	3e-119	58.17%	307	KNA02384.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	274	274	54%	6e-93	71.79%	195	VUZ99735.1
hypothetical protein PVIIG_06431, partial [Plasmodium vivax India VII]	Plasmodium vivax India VII	267	267	70%	6e-90	65.24%	210	KMZ77291.1
VIR protein [Plasmodium vivax]	Plasmodium vivax	189	189	99%	5e-58	41.47%	301	SCA81956.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	177	177	88%	4e-54	41.13%	264	VUZ99426.1
VIR protein [Plasmodium vivax]	Plasmodium vivax	160	160	82%	2e-47	42.11%	248	SCA83617.1
unnamed protein product [Plasmodium vivax]	Plasmodium vivax	157	157	99%	1e-45	37.58%	301	CAG9474536.1
hypothetical protein PVBG_04826 [Plasmodium vivax Brazil I]	Plasmodium vivax Brazil I	149	149	78%	3e-43	40.51%	264	KMZ88617.1
PIR protein [Plasmodium vivax]	Plasmodium vivax	135	135	99%	2e-37	33.12%	292	VUZ99362.1
hypothetical protein PVMG_05324 [Plasmodium vivax Mauritania I]	Plasmodium vivax Mauritania I	129	129	81%	2e-35	34.15%	266	KMZ95725.1
hypothetical protein PVNG_03824 [Plasmodium vivax North Korean]	Plasmodium vivax North Korean	129	129	79%	2e-35	35.56%	266	KMZ98985.1

Next “Edit and resubmit” and change from BLASTP to DELTA-BLAST.



Here is the DELTA-BLAST result:





Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	Select for PSI blast	Used to build PSSM	Newly added
vir1 [Plasmodium vivax]	Plasmodium vivax	363	363	100%	2e-126	100.00%	299	CAB96690.1			
Plasmodium vivax Vir protein, putative [Plasmodium vivax]	Plasmodium vivax	330	330	100%	2e-113	88.63%	299	SCA82045.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	327	327	100%	4e-112	84.62%	299	CAI7724112.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	314	314	100%	5e-107	85.28%	298	CAI7719847.1			
VIR protein [Plasmodium vivax]	Plasmodium vivax	309	309	100%	6e-105	80.27%	299	SCA60268.1			
VIR protein [Plasmodium vivax]	Plasmodium vivax	291	291	100%	4e-98	82.27%	299	SCA60058.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	284	284	100%	3e-95	83.61%	299	VUZ93672.1			
unnamed protein product [Plasmodium vivax]	Plasmodium vivax	270	270	100%	1e-89	66.22%	299	CAG9473090.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	266	266	100%	3e-88	72.58%	298	VUZ99642.1			
unnamed protein product [Plasmodium vivax]	Plasmodium vivax	265	265	100%	6e-88	69.90%	299	CAG9485322.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	247	247	100%	4e-80	58.38%	334	VUZ99734.1			
hypothetical protein PVNG_01538 [Plasmodium vivax North Korean]	Plasmodium vivax North Kor...	218	218	97%	2e-69	57.84%	307	KNA02384.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	196	196	54%	4e-62	71.79%	195	VUZ99735.1			
hypothetical protein PVMG_05816 [Plasmodium vivax Mauritania I]	Plasmodium vivax Mauritania I	194	194	79%	6e-61	83.47%	244	KMZ95672.1			
VIR protein [Plasmodium vivax]	Plasmodium vivax	150	150	99%	4e-43	40.74%	301	SCA81956.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	138	138	88%	7e-39	40.30%	264	VUZ99426.1			
hypothetical protein PVIIG_06431, partial [Plasmodium vivax India VII]	Plasmodium vivax India VII	132	132	70%	3e-37	65.24%	210	KMZ77291.1			
hypothetical protein PVNG_06217 [Plasmodium vivax North Korean]	Plasmodium vivax North Kor...	125	125	49%	3e-32	23.12%	516	KNA01516.1			
PIR protein [Plasmodium vivax]	Plasmodium vivax	124	124	61%	4e-32	14.29%	391	VUZ97285.1			
unnamed protein product [Plasmodium vivax]	Plasmodium vivax	124	124	61%	4e-32	16.84%	405	CAG9473105.1			
VIR protein, partial [Plasmodium vivax]	Plasmodium vivax	124	124	49%	7e-32	22.64%	455	SCA60244.1			

In a way the results of this particular search are in some sense surprising. BLASTP successfully found a series of matches with very low E values. Whereas DELTA-BLAST successfully found many more matches (using its PSSM-based approach). However its best match (2e-126) does not have as low an Expect value as produced by

BLASTP. In many other DELTA-BLAST searches the E values are better (lower) than those found by BLASTP. The result shown here follows from the PSSM approach used by DELTA-BLAST in which a large set of vir1-related proteins were aligned and scored to define the general properties of this family.

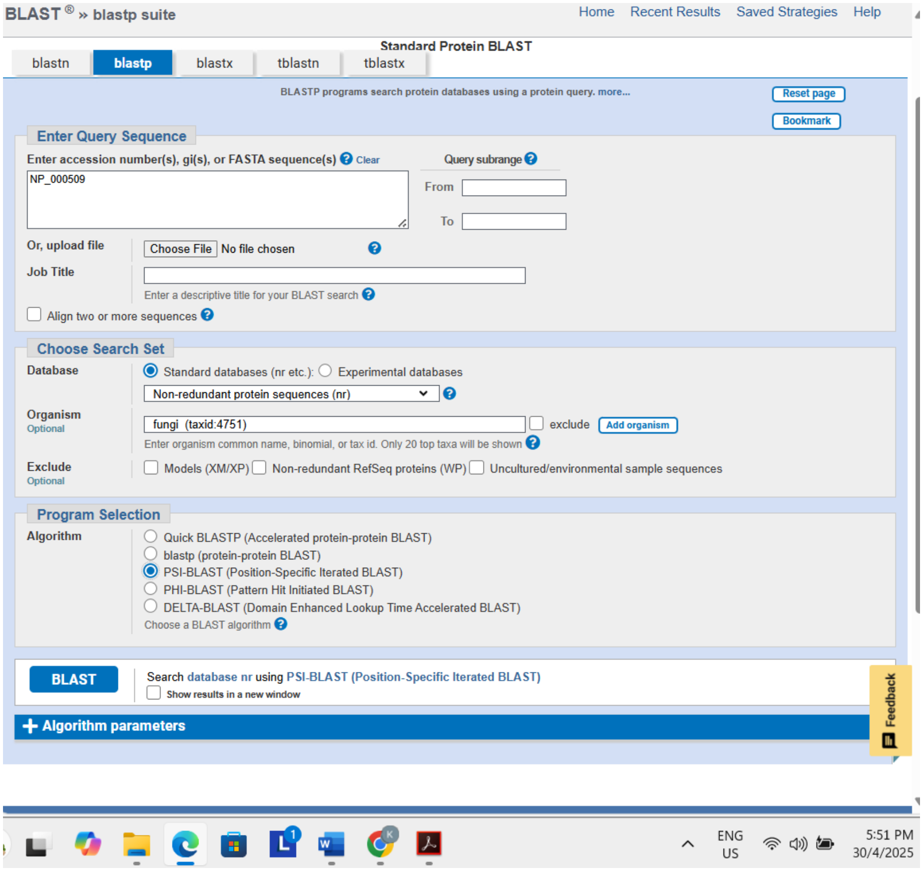
3) Are there globins in fungi?

i) Perform a PSI-BLAST search using human beta globin (NP_000509) as a query, restricting the output to sequences from fungi (taxid:4751) in the nr database.

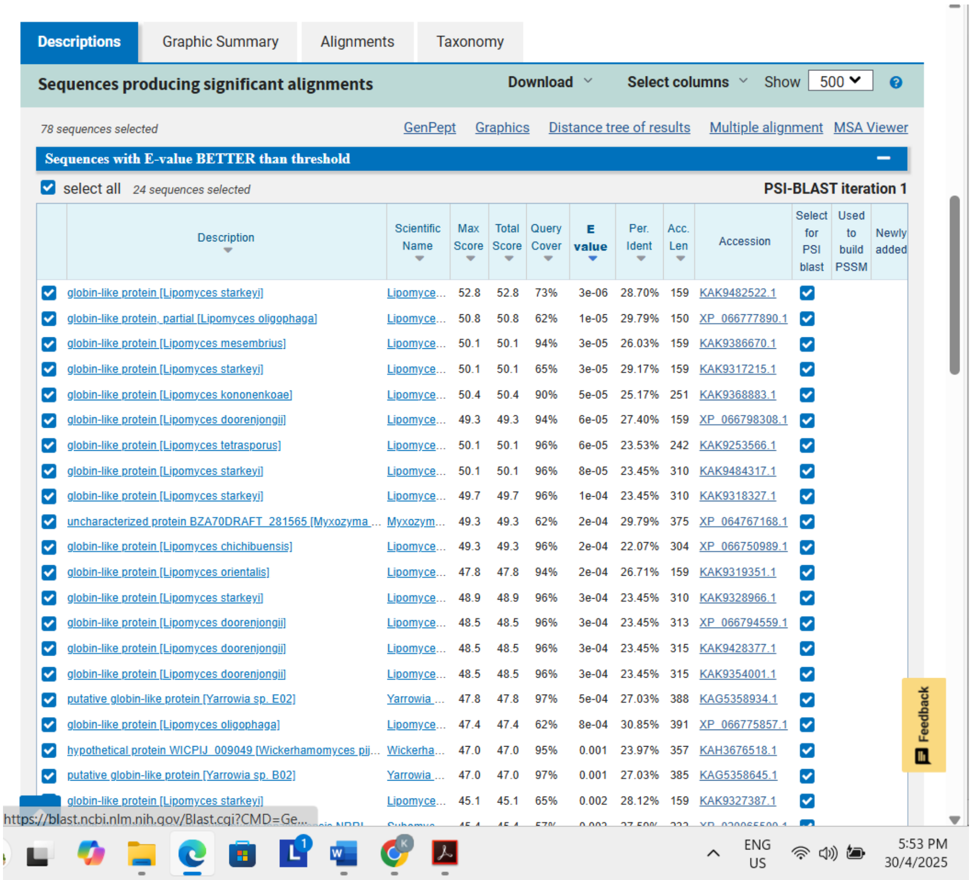
ii) What is the approximate range of lengths of fungal proteins having globin domains? What non-globin domains are often present in fungal globins? Does the presence of these unrelated domains lead to corruption? Why or why not?

iii) In the first iteration there are several hits (with the E values below the 0.005 threshold). After several more iterations there are many dozens of hits including flavohemoproteins that include a globin domain. These fungal proteins have globin domains that are more related to bacterial than vertebrate orthologs. Most of the fungal flavohemoproteins and are quite long (over 400 amino acids and sometimes about 1000 amino acids long), having multiple domains. However, only the globin domain is used for the continued PSI-BLAST iterations.

Lets perform a PSI-BLAST search using human beta globin (NP_000509) as a query, restricting the output to sequences from fungi (taxid:4751) in the nr database.



The result that there are indeed many globins in fungi. Here show the PSI-BLAST results:


Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments — Download — Select columns — Show 500

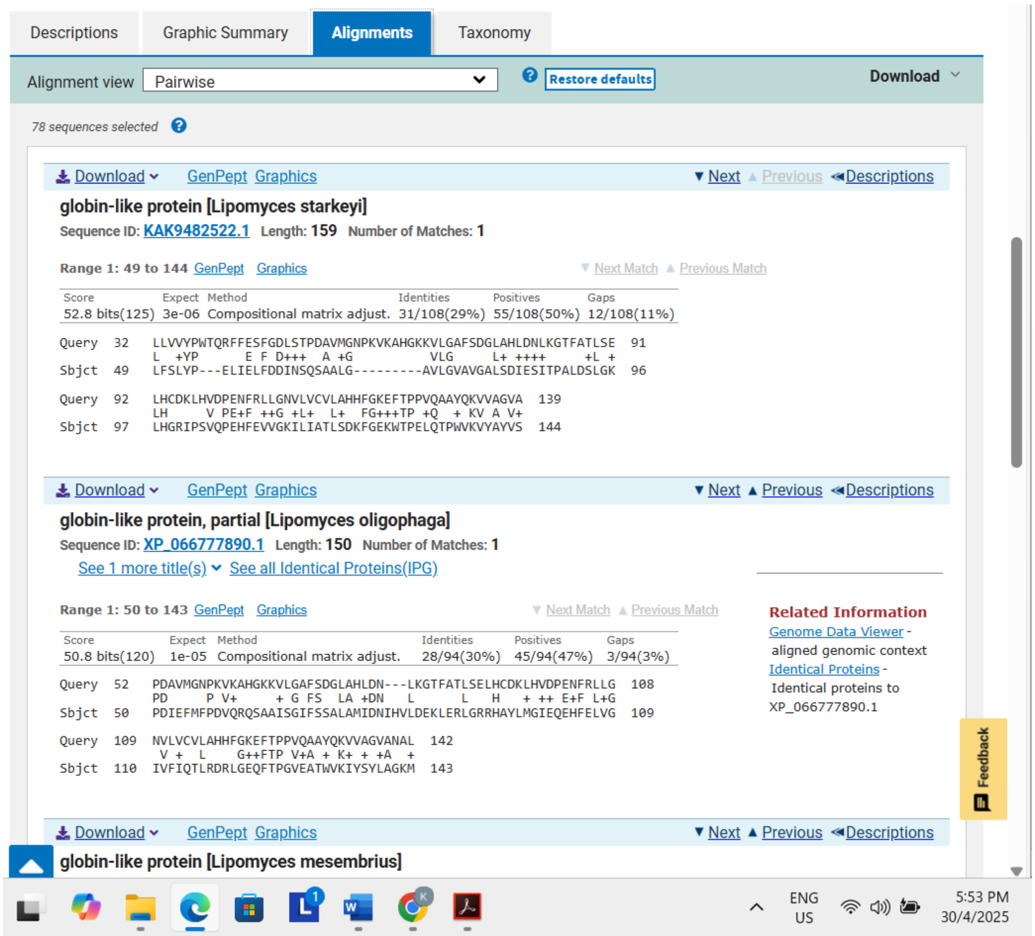
78 sequences selected — GenPept | Graphics | Distance tree of results | Multiple alignment | MSA Viewer

Sequences with E-value BETTER than threshold

select all 24 sequences selected — PSI-BLAST iteration 1

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	Select for PSI blast	Used to build PSSM	Newly added
globin-like protein [Lipomyces starkeyi]	Lipomyce...	52.8	52.8	73%	3e-06	28.70%	159	KAK9482522.1	☑		
globin-like protein, partial [Lipomyces oligophaga]	Lipomyce...	50.8	50.8	62%	1e-05	29.79%	150	XP_066777890.1	☑		
globin-like protein [Lipomyces mesembrius]	Lipomyce...	50.1	50.1	94%	3e-05	26.03%	159	KAK9386670.1	☑		
globin-like protein [Lipomyces starkeyi]	Lipomyce...	50.1	50.1	65%	3e-05	29.17%	159	KAK9317215.1	☑		
globin-like protein [Lipomyces kononenkoae]	Lipomyce...	50.4	50.4	90%	5e-05	25.17%	251	KAK9368883.1	☑		
globin-like protein [Lipomyces doorenjongii]	Lipomyce...	49.3	49.3	94%	6e-05	27.40%	159	XP_066798308.1	☑		
globin-like protein [Lipomyces tetrasporus]	Lipomyce...	50.1	50.1	96%	6e-05	23.53%	242	KAK9253566.1	☑		
globin-like protein [Lipomyces starkeyi]	Lipomyce...	50.1	50.1	96%	8e-05	23.45%	310	KAK9484317.1	☑		
globin-like protein [Lipomyces starkeyi]	Lipomyce...	49.7	49.7	96%	1e-04	23.45%	310	KAK9318327.1	☑		
uncharacterized protein BZA70DRAFT_281565 [Myxozyma ...	Myxozym...	49.3	49.3	62%	2e-04	29.79%	375	XP_064767168.1	☑		
globin-like protein [Lipomyces chichibuensis]	Lipomyce...	49.3	49.3	96%	2e-04	22.07%	304	XP_066750989.1	☑		
globin-like protein [Lipomyces orientalis]	Lipomyce...	47.8	47.8	94%	2e-04	26.71%	159	KAK9319351.1	☑		
globin-like protein [Lipomyces starkeyi]	Lipomyce...	48.9	48.9	96%	3e-04	23.45%	310	KAK9328966.1	☑		
globin-like protein [Lipomyces doorenjongii]	Lipomyce...	48.5	48.5	96%	3e-04	23.45%	313	XP_066794559.1	☑		
globin-like protein [Lipomyces doorenjongii]	Lipomyce...	48.5	48.5	96%	3e-04	23.45%	315	KAK9428377.1	☑		
globin-like protein [Lipomyces doorenjongii]	Lipomyce...	48.5	48.5	96%	3e-04	23.45%	315	KAK9354001.1	☑		
putative globin-like protein [Yarrowia sp. E02]	Yarrowia ...	47.8	47.8	97%	5e-04	27.03%	388	KAG5358934.1	☑		
globin-like protein [Lipomyces oligophaga]	Lipomyce...	47.4	47.4	62%	8e-04	30.85%	391	XP_066775857.1	☑		
hypothetical protein WICPIJ_009049 [Wickerhamomyces pij...	Wickerha...	47.0	47.0	95%	0.001	23.97%	357	KAH3676518.1	☑		
putative globin-like protein [Yarrowia sp. B02]	Yarrowia ...	47.0	47.0	97%	0.001	27.03%	385	KAG5358645.1	☑		
globin-like protein [Lipomyces starkeyi]	Lipomyce...	45.1	45.1	65%	0.002	28.12%	159	KAK9327387.1	☑		

To see the lengths of these matches inspect the pairwise outputs. You can also reformat to view (or export) the resuts as a table. Most fungal proteins have lengths of 250 to 450 amino acids. Some are larger, e.g.:



In many cases these larger fungal globins contain repeating globin units. Most fungal proteins have lengths of 250 to 450 amino acids. S globin are often present in fungal globins. Presence of unrelated domains can lead to corruption which result in the high scores for homologous to the unrelated domains. This corruption can then lead to high scores for the assignment and statistical significance to biological-incorrect relationship.

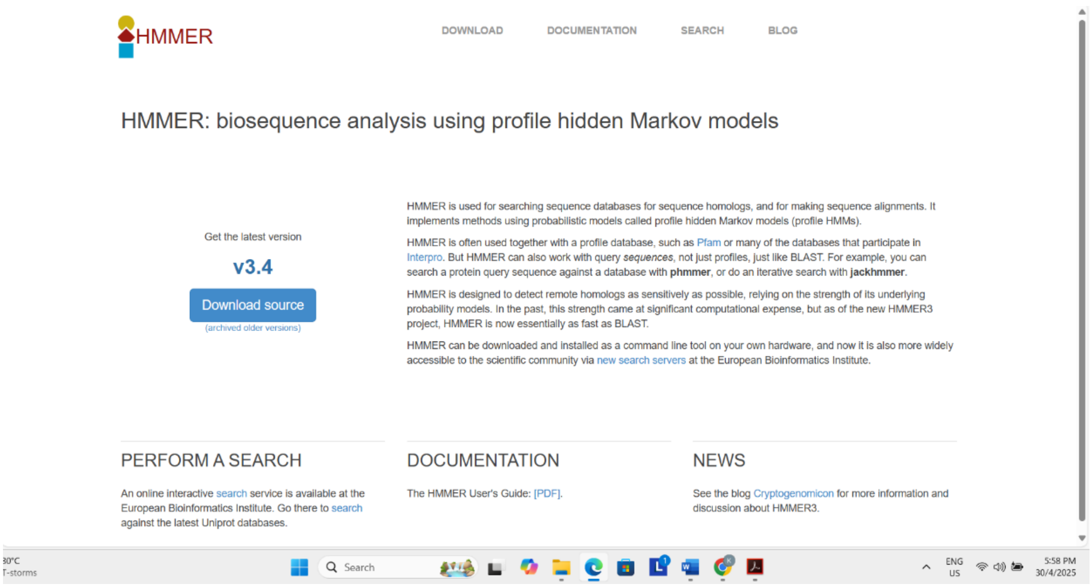
4) Perform HMMER searches

i) First make two different HMMs. You can obtain sets of vertebrate globin and bacterial/fungal/vertebrate globin sequences as web documents 5.6 and 5.7. The multiple sequence alignments that we use as input to HMMER are in these documents.

ii) When the profile HMM was built from a multiple sequence alignment of vertebrate alpha and beta globins and used to search the human RefSeq database, there were many database matches, including myoglobin that we could not detect with BLASTP. In contrast, when an alignment of bacterial and fungal globins was used to generate a profile HMM, the output consisted of one result with a non-signficant expect value. Combining several human globins with the bacterial and fungal globins in a multiple sequence alignment resulted in the creation of an HMM that readily detected human globins. Thus, the profile HMM is a model that is sensitive to the choice of sequences that are used as input for the multiple sequence alignment.

iii) The full results of the HMMER searches for (1) vertebrate, (2) bacterial plus fungal, and (3) bacterial plus fungal plus vertebrate globins are shown in web documents 5.8, 5.9, and 5.10. The HMM match to human myoglobin had a higher score and lower E value in search (3) than in (1).

iv) HMMer searches are run locally. This search was run against all human RefSeq proteins. You can download NCBI databases such as RefSeq by visiting the file transfer protocol (FTP) site from the home page of NCBI or going directly to ►http://www.ncbi.nlm.nih.gov/Ftp/ (WebLink 5.1). Place the downloaded database into the same directory as your input sequences for HMMER.



Download hmmer software

Create a new directory (mkdir hmmer)

Change the directory to hmmer: cd hmmer

Move the hmmer package in the download directory to this new created directory :

mv ~/Downloads/hmm* hmmer

```
Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https:/
/aka.ms/PSWindows

PS C:\Users\user> mkdir hmmer
mkdir : An item with the specified name C:\Users\user\hmmer already
exists.
At line:1 char:1
+ mkdir hmmer
+ ~~~~~~~~~~~
    + CategoryInfo          : ResourceExists: (C:\Users\user\hmmer:Str
   ing) [New-Item], IOException
    + FullyQualifiedErrorId : DirectoryExist,Microsoft.PowerShell.Comm
   ands.NewItemCommand

PS C:\Users\user> cd hmmer
PS C:\Users\user\hmmer> mv ~/Downloads/hmm* hmmer
PS C:\Users\user\hmmer> |
```
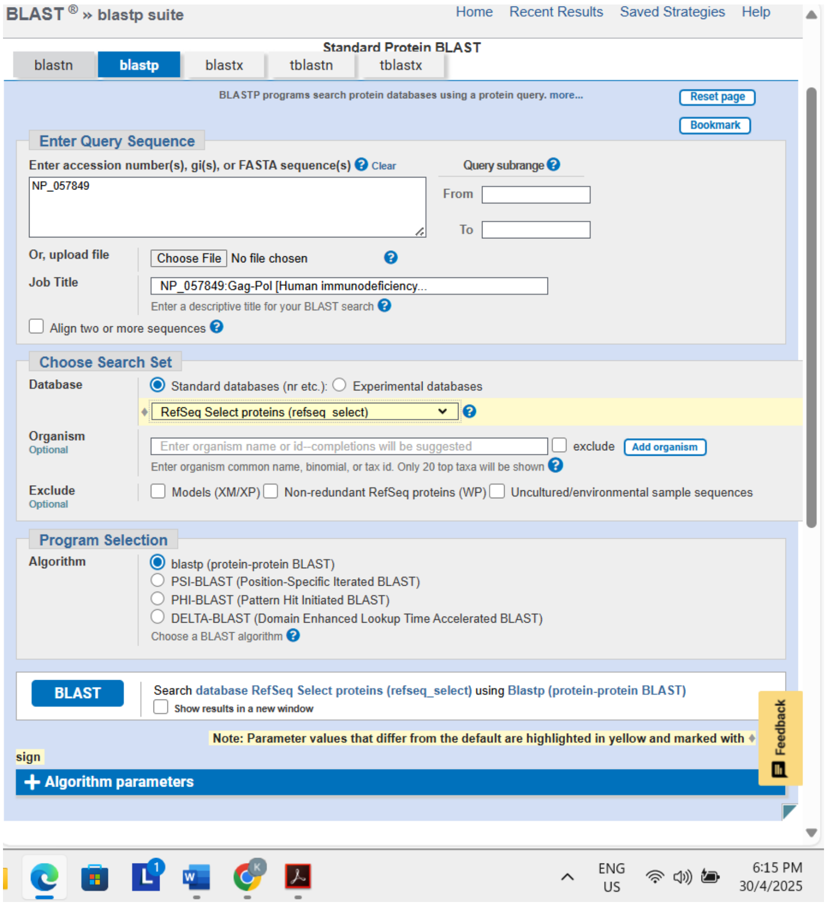
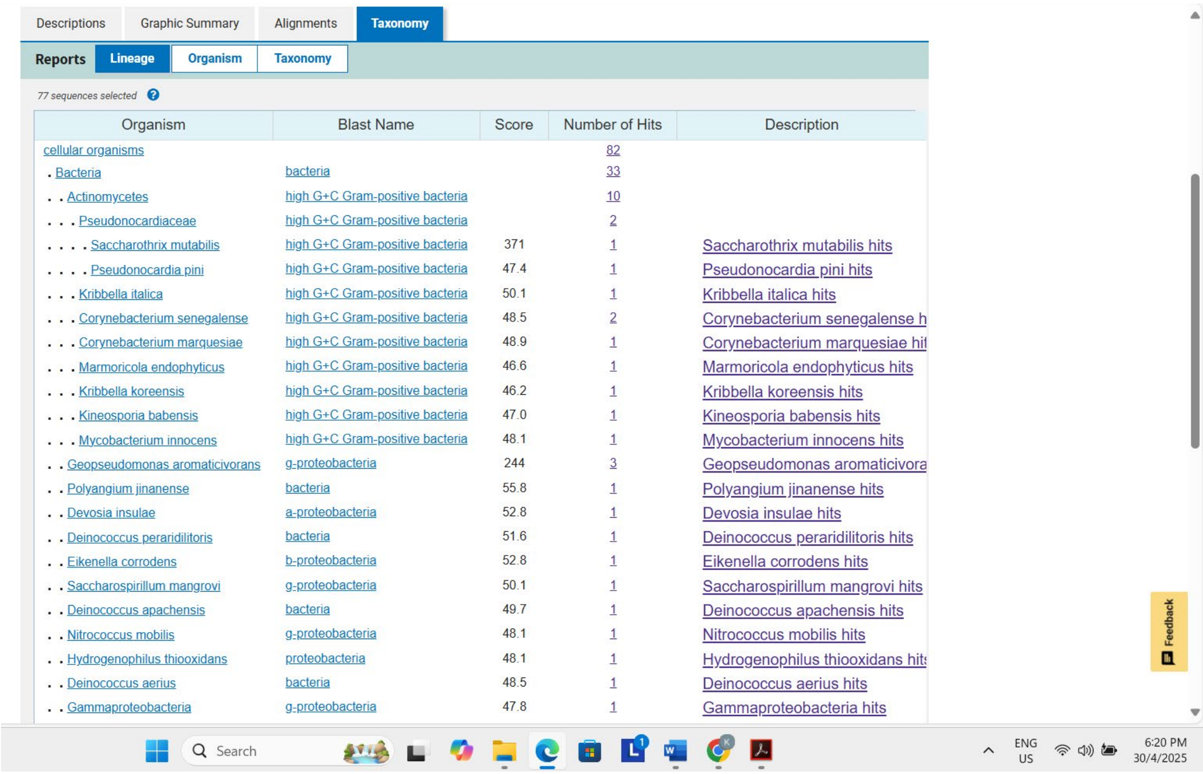

5) We previously performed a series of BLAST searches using HIV-1 pol as a query (NP_057849).

   i) Perform a BLASTP search using this query. Look at the taxonomy report to see which viruses match this query.
   ii) Next, repeat the search using several iterations of PSI-BLAST. Compare this taxonomy report to that of the BLASTP search.
   iii) What do you observe? Are there any nonviral sequences detected in the PSI-BLAST search?
   iv) Did you expect to find any?

BLASTP searches using HIV-1 pol as a query (NP_057849).

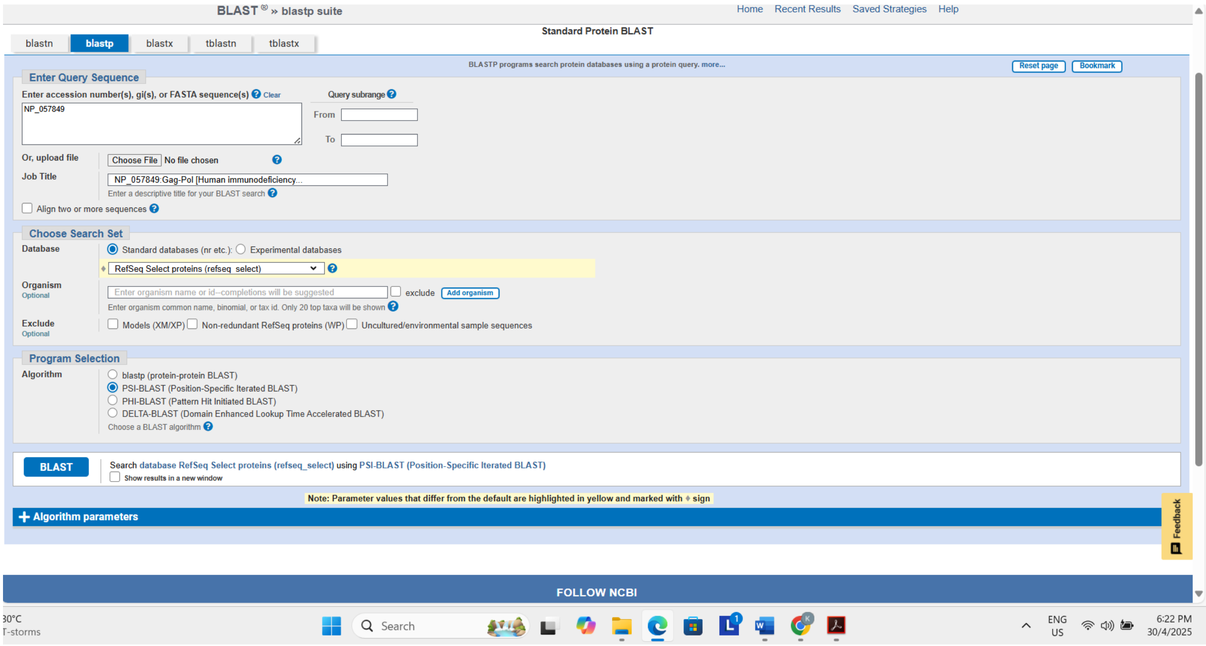


This shows the taxonomy report for BLASTP for NP_057849.

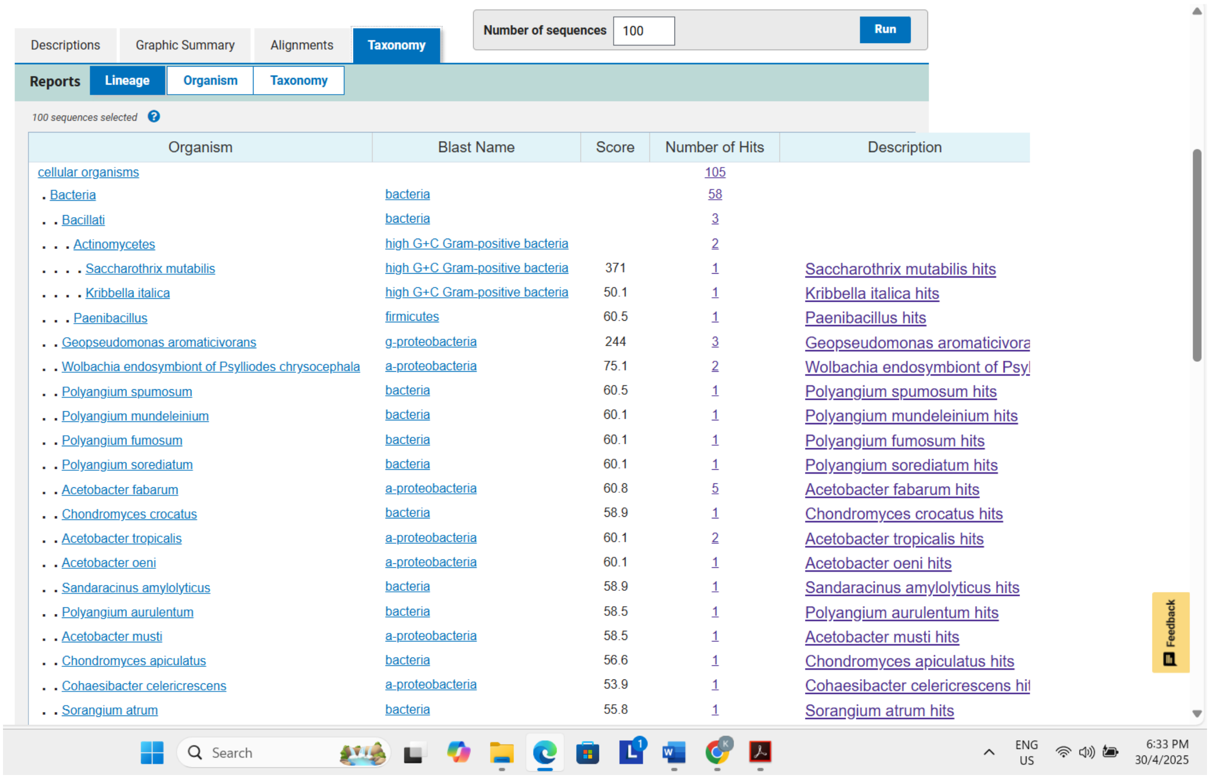


Organism	Blast Name	Score	Number of Hits	Description
cellular organisms			82	
. Bacteria	bacteria		33	
. . Actinomycetes	high G+C Gram-positive bacteria		10	
. . . Pseudonocardiaceae	high G+C Gram-positive bacteria		2	
. . . . Saccharothrix mutabilis	high G+C Gram-positive bacteria	371	1	Saccharothrix mutabilis hits
. . . . Pseudonocardia pini	high G+C Gram-positive bacteria	47.4	1	Pseudonocardia pini hits
. . . Kribbella italica	high G+C Gram-positive bacteria	50.1	1	Kribbella italica hits
. . . Corynebacterium senegalense	high G+C Gram-positive bacteria	48.5	2	Corynebacterium senegalense h
. . . Corynebacterium marquesiae	high G+C Gram-positive bacteria	48.9	1	Corynebacterium marquesiae hi
. . . Marmoricola endophyticus	high G+C Gram-positive bacteria	46.6	1	Marmoricola endophyticus hits
. . . Kribbella koreensis	high G+C Gram-positive bacteria	46.2	1	Kribbella koreensis hits
. . . Kineosporia babensis	high G+C Gram-positive bacteria	47.0	1	Kineosporia babensis hits
. . . Mycobacterium innocens	high G+C Gram-positive bacteria	48.1	1	Mycobacterium innocens hits
. . Geopseudomonas aromaticivorans	g-proteobacteria	244	3	Geopseudomonas aromaticivora
. . Polyangium jinanense	bacteria	55.8	1	Polyangium jinanense hits
. . Devosia insulae	a-proteobacteria	52.8	1	Devosia insulae hits
. . Deinococcus peraridilitoris	bacteria	51.6	1	Deinococcus peraridilitoris hits
. . Eikenella corrodens	b-proteobacteria	52.8	1	Eikenella corrodens hits
. . Saccharospirillum mangrovi	g-proteobacteria	50.1	1	Saccharospirillum mangrovi hits
. . Deinococcus apachensis	bacteria	49.7	1	Deinococcus apachensis hits
. . Nitrococcus mobilis	g-proteobacteria	48.1	1	Nitrococcus mobilis hits
. . Hydrogenophilus thiooxidans	proteobacteria	48.1	1	Hydrogenophilus thiooxidans hit
. . Deinococcus aerius	bacteria	48.5	1	Deinococcus aerius hits
. . Gammaproteobacteria	g-proteobacteria	47.8	1	Gammaproteobacteria hits

PSI-BLAST searches using HIV-1 pol as a query (NP_057849).

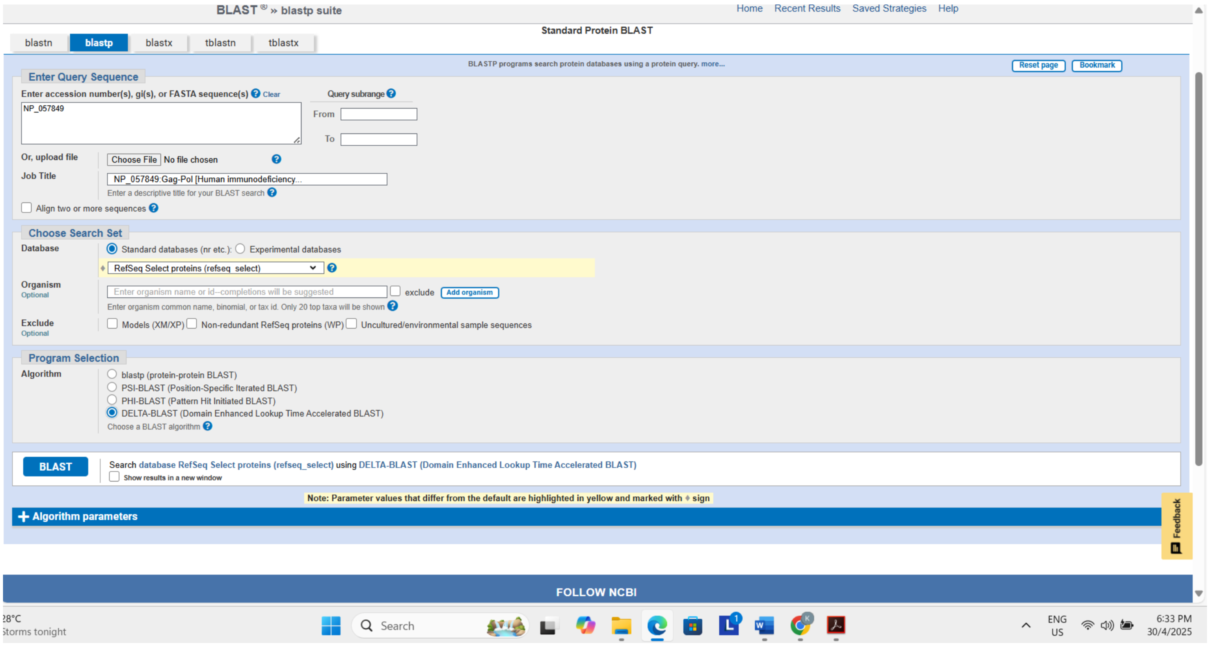


This shows the taxonomy report for PSI-BLAST for NP_057849.

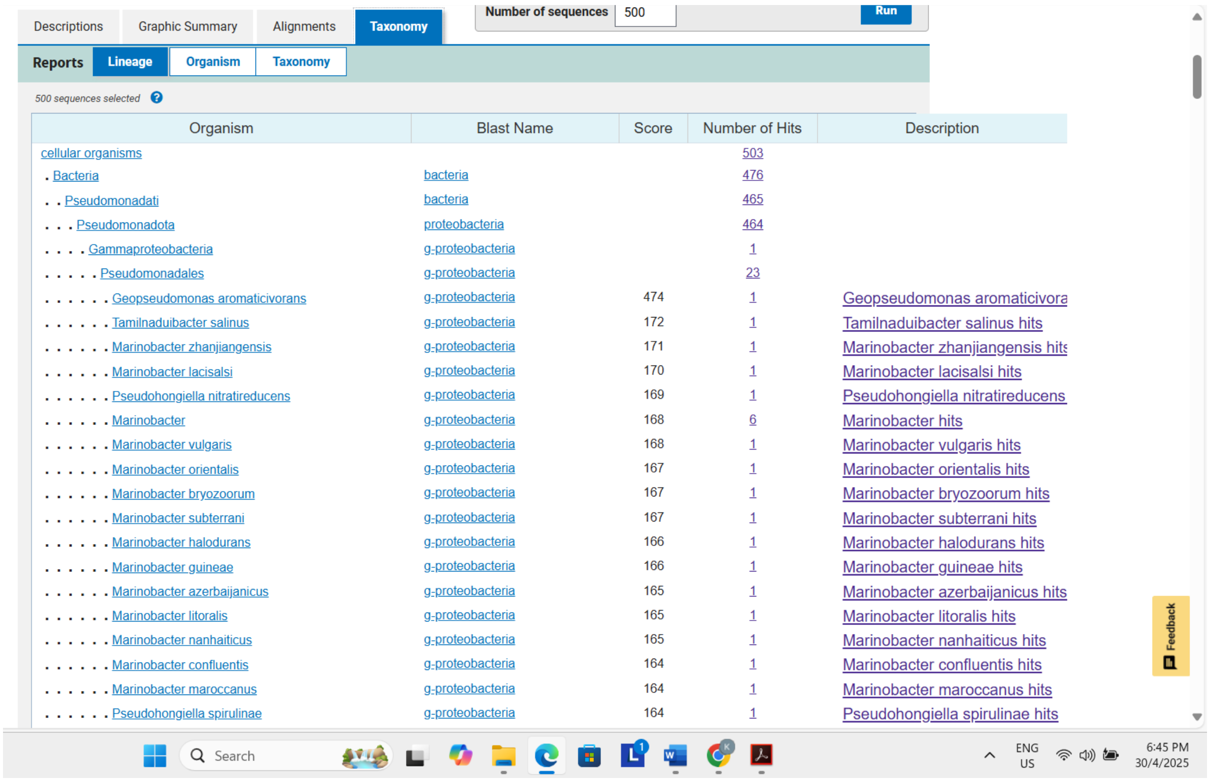


Organism	Blast Name	Score	Number of Hits	Description
cellular organisms			105	
. Bacteria	bacteria		58	
. . Bacillati	bacteria		3	
. . . Actinomycetes	high G+C Gram-positive bacteria		2	
. . . . Saccharothrix mutabilis	high G+C Gram-positive bacteria	371	1	Saccharothrix mutabilis hits
. . . . Kribbella italica	high G+C Gram-positive bacteria	50.1	1	Kribbella italica hits
. . . Paenibacillus	firmicutes	60.5	1	Paenibacillus hits
. . Geopseudomonas aromaticivorans	g-proteobacteria	244	3	Geopseudomonas aromaticivora
. . Wolbachia endosymbiont of Psylliodes chrysocephala	a-proteobacteria	75.1	2	Wolbachia endosymbiont of Psyl
. . Polyangium spumosum	bacteria	60.5	1	Polyangium spumosum hits
. . Polyangium mundeleinium	bacteria	60.1	1	Polyangium mundeleinium hits
. . Polyangium fumosum	bacteria	60.1	1	Polyangium fumosum hits
. . Polyangium sorediatum	bacteria	60.1	1	Polyangium sorediatum hits
. . Acetobacter fabarum	a-proteobacteria	60.8	5	Acetobacter fabarum hits
. . Chondromyces crocatus	bacteria	58.9	1	Chondromyces crocatus hits
. . Acetobacter tropicalis	a-proteobacteria	60.1	2	Acetobacter tropicalis hits
. . Acetobacter oeni	a-proteobacteria	60.1	1	Acetobacter oeni hits
. . Sandaracinus amylolyticus	bacteria	58.9	1	Sandaracinus amylolyticus hits
. . Polyangium aurulentum	bacteria	58.5	1	Polyangium aurulentum hits
. . Acetobacter musti	a-proteobacteria	58.5	1	Acetobacter musti hits
. . Chondromyces apiculatus	bacteria	56.6	1	Chondromyces apiculatus hits
. . Cohaesibacter celericrescens	a-proteobacteria	53.9	1	Cohaesibacter celericrescens hi
. . Sorangium atrum	bacteria	55.8	1	Sorangium atrum hits

DELTA-BLAST searches using HIV-1 pol as a query (NP_057849).



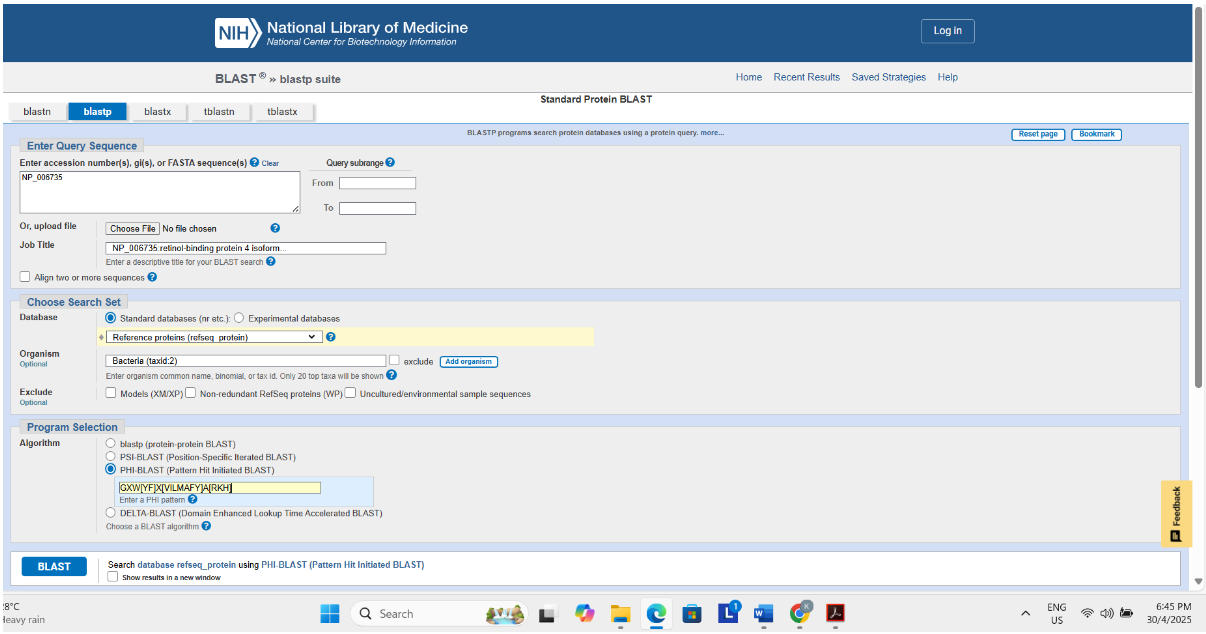
This shows the taxonomy report for DELTA-BLAST for NP_057849.



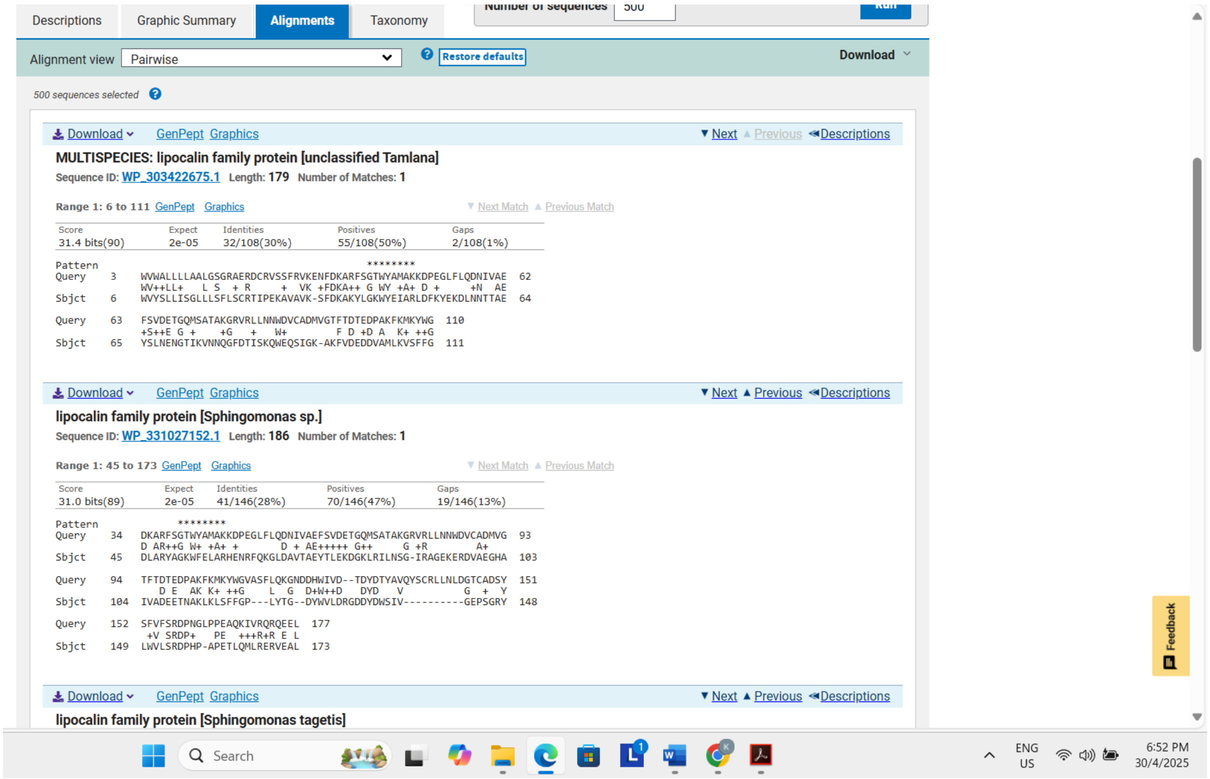
From the results show above, there are more hits with DELTA-BLAST than BLASTP and PSI-BLAST. There are both viral and non-viral detected in the PSI search. This is expected because the PSSM-based approach is far more sensitive than that of standard BLAST.

6) Explore PHI-BLAST using human RBP4 (NP_006735) as a query, restricting the output to bacteria and the RefSeq database.
   i) Use the PHI pattern GXW[YF]X[VILMAFY]A[RKH]. Perform this search, and save the results.
   ii) Then repeat the search using the PHI pattern GXW[YF][EA][IVLM].
   iii) How do the results differ?
   iv) Select one protein that appears as a bacterial protein in a pairwise alignment with the human RBP4 query; what are the E values, and why do they differ?

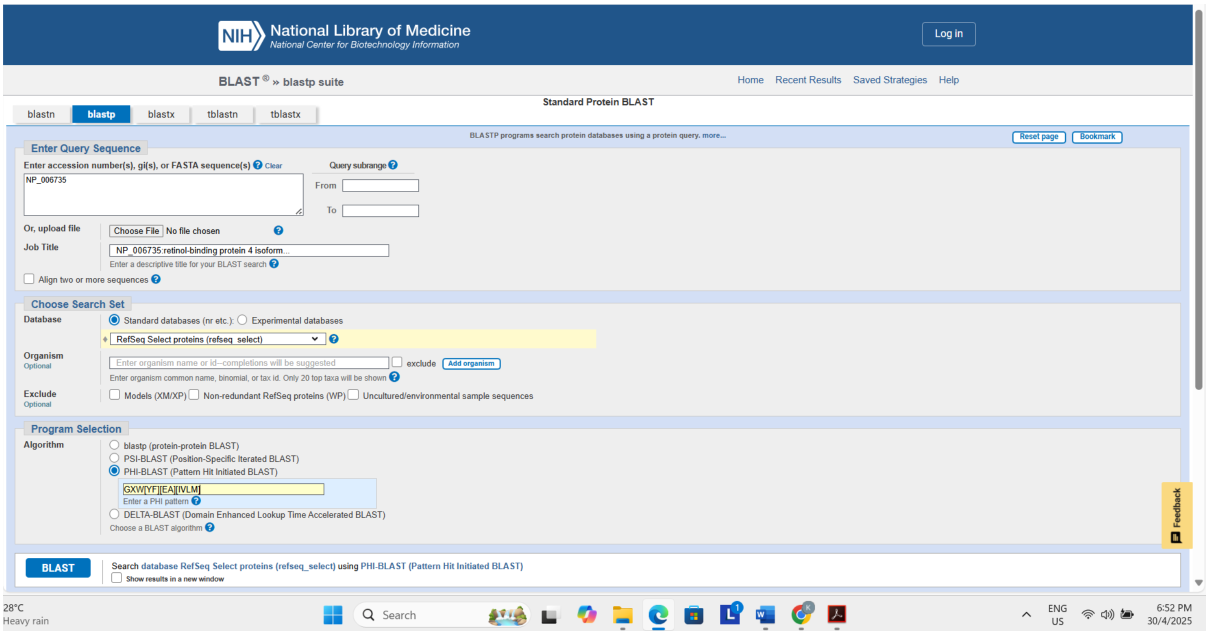
Here is the set-up of the PHI-BLAST using the PHI pattern GXW[YF]X[VILMAFY]A[RKH] search:



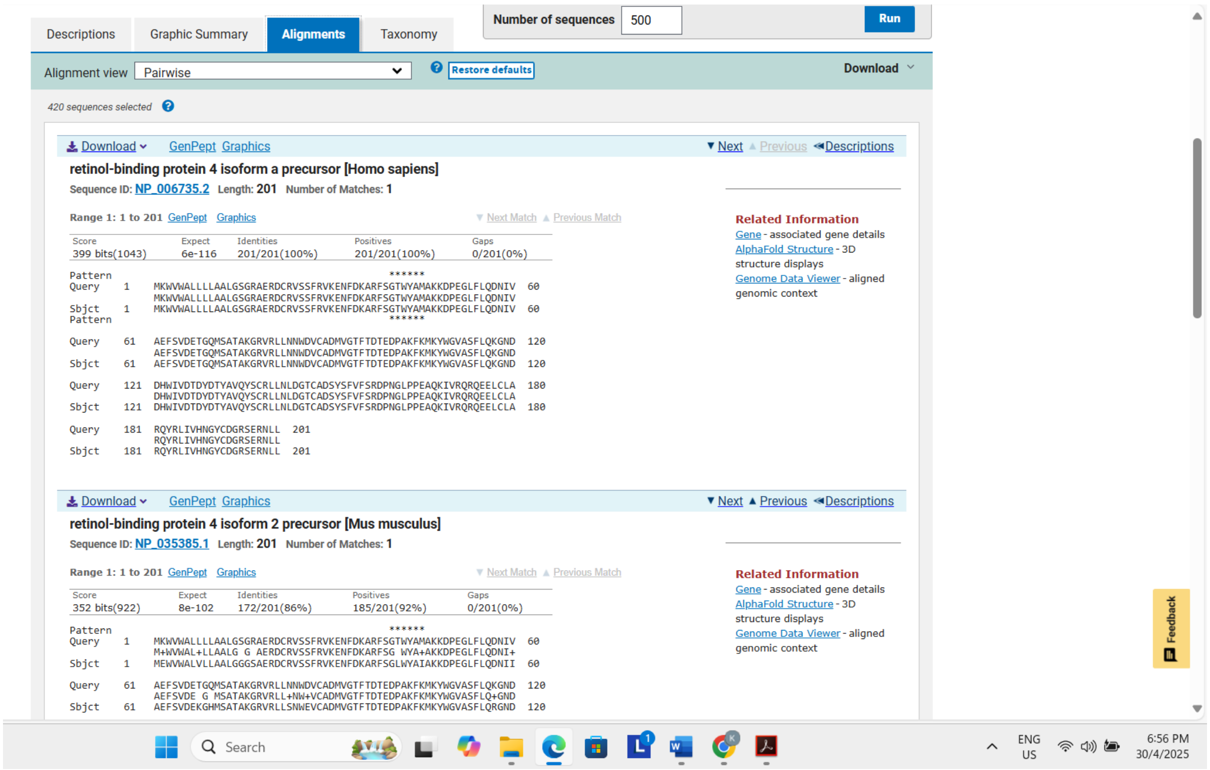
For every database match, the results include asterisks indicating where this pattern matches. And the top two patterns result for PHI-BLAST using the PHI pattern GXW[YF]X[VILMAFY]A[RKH] are shown here:



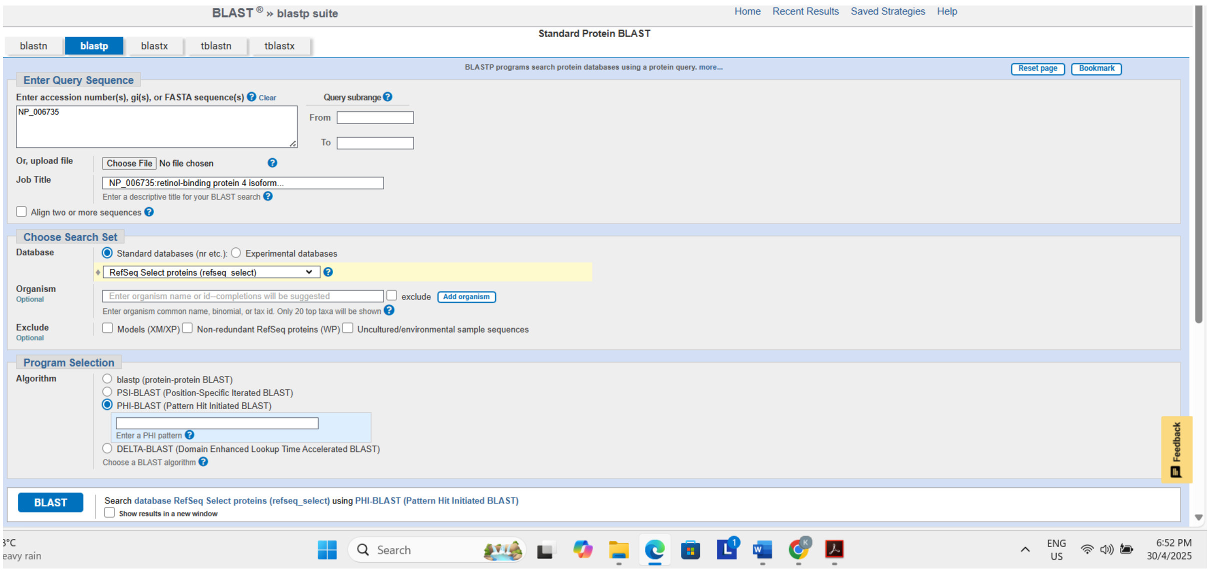
Then repeat the PHI-BLAST using the PHI pattern GXW[YF][EA][IVLM] search:



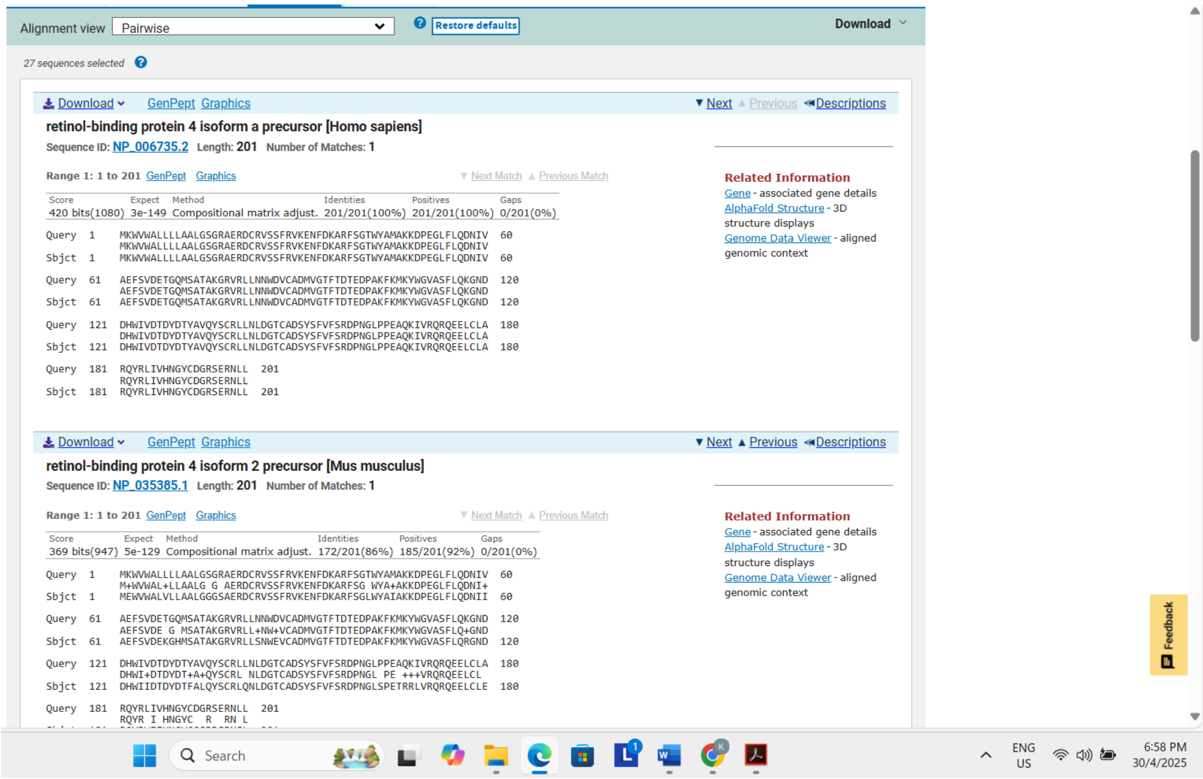
The top two results are identical, except that there are 6 asterisks now instead of 8 asterisks previously. Here the result for PHI-BLAST using the PHI pattern GXW[YF][EA][IVLM]:



Repeat the PHI-BLAST without the PHI pattern search:



Here the result for PHI-BLAST without using the PHI pattern:



Inspecting the results from both searches, the first one (with the longer PHI pattern) had more database matches that were significant (above threshold). If we instead do a PHI-BLAST search without using a PHI pattern, only 27 match is significant. Therefore a PHI pattern can dramatically improve the sensitivity of a BLAST protein search.

Protein that appears as a bacterial protein in a pairwise alignment with the human RBP4 query: Select seq ref|NP_006735.2| retinol-binding protein 4 isoform a precursor [Homo sapiens]

E values:

PHI- BLAST pattern GXW[YF]X[VILMAFY]A[RKH] – 9e-116

PHI- BLAST pattern GXW[YF][EA][IVLM] – 6e-116

PHI- BLAST without pattern – 3e-149

Inspecting the results from three searches, the first one (with the longer PHI pattern) had more database matches that were significant (above threshold). If we instead do a PHI-BLAST search without using a PHI pattern, only 27 match is significant. Therefore a PHI pattern can dramatically improve the sensitivity of a BLAST protein search